2026 4

| 22344140

1

2

3 SETI 1960–2025

4 SETI

5 SKA

6

SYSU

1

2

3 SETI 1960–2025

4 SETI

5 SKA

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7

SYSU

- 1959 Cocconi Morrison *Nature*
- 1960 Frank Drake SETI
- technosignature

SYSU

-
-
-
- SKA SETI

Definition 1.1

SETI

SYSU

-
-
- SKA SETI
- SETI

1

2

3 SETI 1960–2025

4 SETI

5 SKA

6

7

SYSU

- **ADS** 1960–2025 SETI
- Cocconi-Morrison technosignature
- Breakthrough Listen 1327 1.10–3.45 GHz
- **SKAO**

SYSU

- / SETI technosignature → **778**
- → **1519**
-

1. —— deg^2

- $N_{\mathrm{tgt}} \times \Omega_b$
-

2. —— GHz

- $\Delta f = f_{\mathrm{max}} - f_{\mathrm{min}}$
- ” ” ” ”

3. ——

- $A_{\mathrm{eff}}/T_{\mathrm{sys}}$ SEFD
- m^2

SYSU

1

2

3 SETI 1960–2025

4 SETI

5 SKA

6

7

SYSU

SETI

1960s–70s

- Cocconi-Morrison 1959 Drake 1960
- 4 $\bar{N} \approx 0.36$ /
-

1971–92

- NASA HRMS
- 381 $\bar{N} \approx 17.3$ /
- $g = 9425\%$

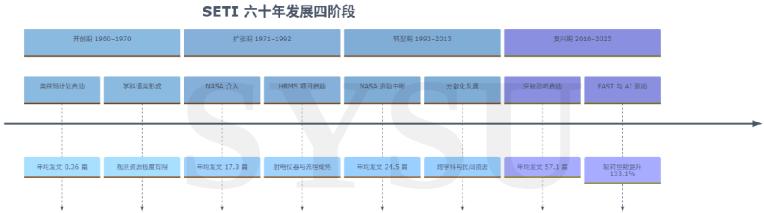
1993–2015

- NASA
- 563 $\bar{N} \approx 24.5$ /
-

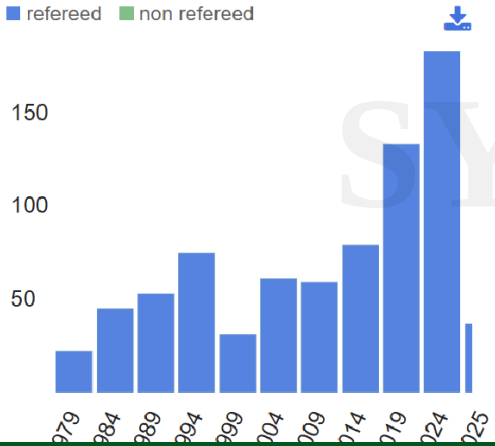
2016–25

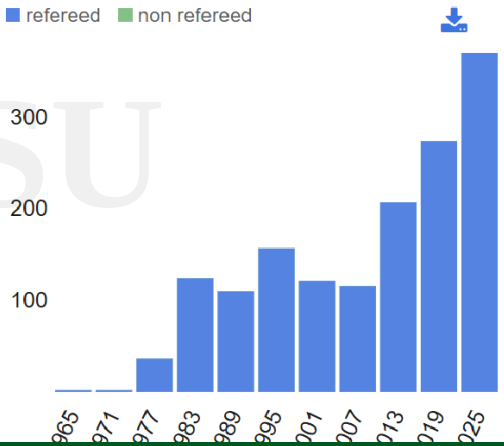
- BL FAST
- 571 $\bar{N} \approx 57.1$ /
- 133%

SYSU

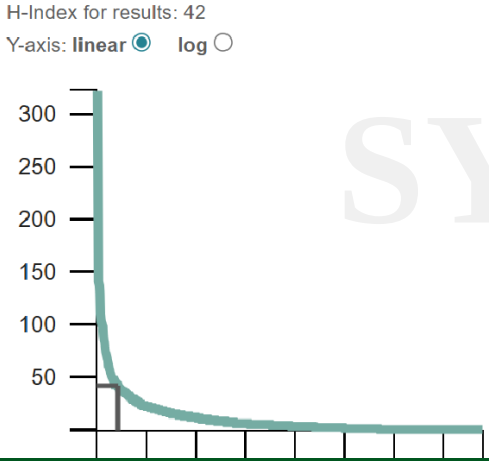

SETI 六十年发展四阶段
开创期 1960–1970
扩张期 1971–1992
转型期 1993–2015
复兴期 2016–2025
奥斯特计划启动
学科雏形形成
NASA 介入
HBMS 项目启动
NASA 退出中期
十年空白期
中期回归启动
FAST 与 AI 崛起
年均发文 0.06 篇
取样资源极度有限
年均发文 17.3 篇
射电仪器与处理模块
年均发文 24.5 篇
跨学科与民间资源
年均发文 57.1 篇
取样资源提升 133.1%

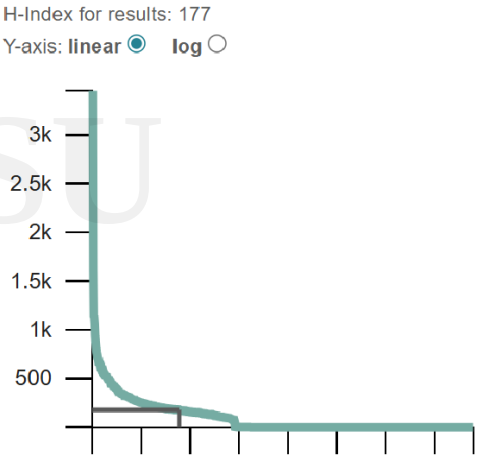
3.1 SETI


refereed
non refereed
150
100
50
979
984
989
994
999
004
009
014
019
024
025


refereed
non refereed
300
200
100
965
971
977
983
989
995
001
007
013
019
025

SYSU


H-Index for results: 42
Y-axis: linear log
300
250
200
150
100
50


H-Index for results: 177
Y-axis: linear log
3k
2.5k
2k
1.5k
1k
500

SYSU

-
-
-

- RFI
- ” ”
-

Definition 3.1

SETI ” ”

SYSU

1
2
3 SETI 1960–2025
4 SETI
5 SKA
6
7

SYSU

SETI

SETI

+ RFI

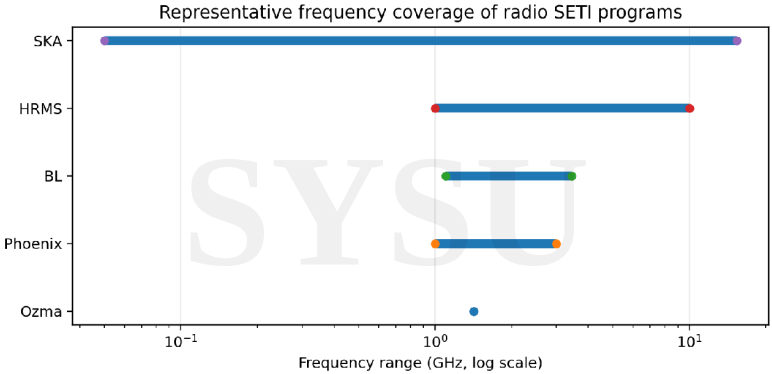
” ”

” ” ” ”

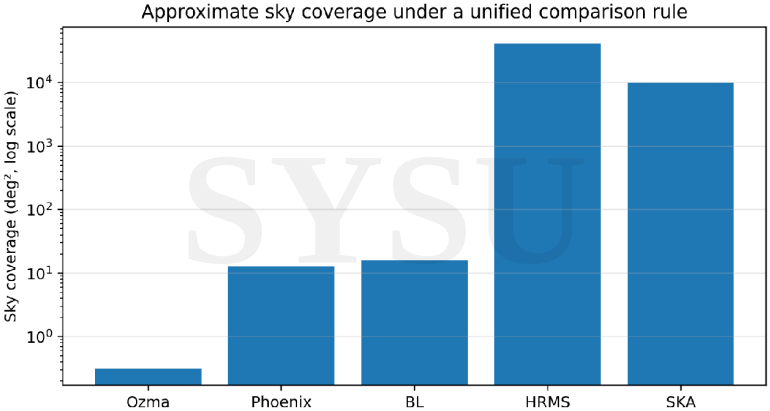
COBE Planck

1
2
3 SETI 1960–2025
4 SETI
5 SKA
6
7

SYSU

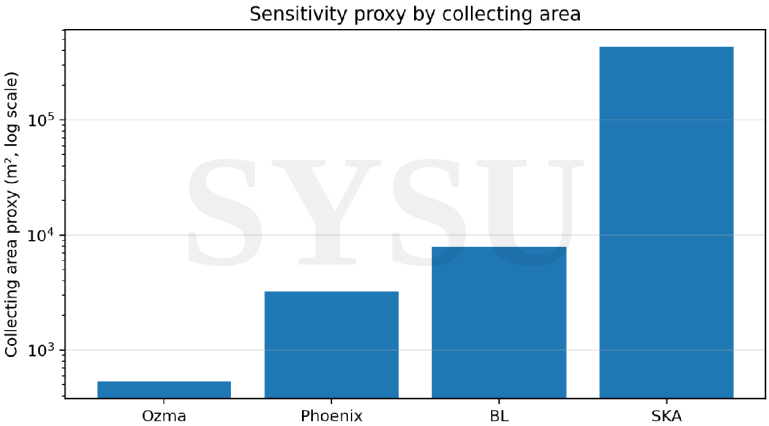

Representative frequency coverage of radio SETI programs
SKA
HRMS
BL
Phoenix
Ozma
10^{-1}
10^{0}
10^{1}
Frequency range (GHz, log scale)

5.1 SETI


Approximate sky coverage under a unified comparison rule
Sky coverage (deg², log scale)
10^4
10^3
10^2
10^1
10^0
Ozma
Phoenix
BL
HRMS
SKA

5.2 SETI

SKA ” ”


Sensitivity proxy by collecting area
Collecting area proxy (m², log scale)
10^5
10^4
10^3
Ozma
Phoenix
BL
SKA

5.3

” ”

Theorem 5.1

SKA ” ”

	$\mathbf{deg}^2$	**GHz**	$\mathbf{m}^2$
	0.31	1.42	∼531
	12.8	1–3	∼3,217
Breakthrough Listen	16.0	1.10–3.45	∼7,854
NASA HRMS	4.1×10^4	1–10	
SKA	$\mathbf{10^3}$**–**$\mathbf{10^4}$	**0.05–15.4**	$\sim \mathbf{4.3 \times 10^5}$

SYSU

1
2
3 SETI 1960–2025
4 SETI
5 SKA
6
7

SYSU

1.

SETI

2.

× × SKA

3. SKA

SKA SETI SETI

4.

SYSU

SETI

SETI

RFI

technosignature

SYSU

1
2
3 SETI 1960–2025
4 SETI
5 SKA
6
7

SYSU

| 22344140